Paper Title: Programming in C and Data Structures Paper Code: CS – 2127

Course Outcome:

After successful completion of course the students will be able to:

- Understand the fundamental concepts of C programming language.
- Familiarize with common data structures such as arrays, linked lists, stacks, queues, trees and graphs.
- Identify and implement appropriate data types and structures for problem-solving using C.
- Learn further through investigation of program behaviour and debugging efforts.

Paper Title: Computer Organization and Architecture Paper Code : CS – 2111

Course outcomes:

After successful completion of course the students will be able to:

- Solve the functions and expression and built various circuits using logic gates.
- Demonstrate computer architecture concepts related to design of modern processors,memories and I/Os.
- Understand the theory and architecture of central processing unit and functionality ofcentral processing unit. I/O and memory organization.
- Understand the concepts of parallel processing, pipelining and inter-processorcommunication.

Paper Title: Mathematical Structures and Linear Programming Paper Code: CS – 2126

Course Outcomes:

After successful completion of course the students will be able to:

- Develop the ability to think, read and write Algorithms effectively. The functionsunderstanding including comparison of functions will help in programming concepts.
- Understand important logical concepts as tautologies, contradictions, Laws in Logic, logicalequivalence, formulating quantified statements, truth tables and truthfulness of a compoundstatement.
- Understand numerical, mathematics contents and solutions to Scientific Problem like solutionto non-linear equations using Bisection method, False position method, Newton-Raphsonmethod, Secant method.
- Understand various methods of interpolation, numerical integration and Linear and Non-Linear
- Regression which will help the students to understand important concepts and methods innumerical analysis.

Paper Title: Relational Database Management Systems Paper Code: CS – 2113

Course Outcomes:

After successful completion of course the students will be able to:
• Describe the fundamental elements of Relational Database Management Systems, basicconcepts of Relational data model, Entity-relationship model, Relational database design,
Relational Algebra SQL.
• Design ER-models and convert the ER-model to relational tables, Normalize database andpopulate relational database.
• Manipulation of databases using SQL commands.
• Understanding block structure of PL/SQL, functions and procedures using PL/SQL andperform Cursor and Trigger Management.

Paper Title: Operating System Paper Code: CS – 2114

Course Outcomes:

After successful completion of course the students will be able to:
• Describe the important computer system resources and the role of operating system in theirmanagement policies and algorithms.
• Understand the process management policies and scheduling of processes by CPU
• Evaluate the requirement for process synchronization and coordination, memory managementhandled by operating system also cater-policies with respect to different storage.
• Learn to use features of Linux for programming and administration

Paper Title: Object-Oriented Programming using C++ and Java Paper Code : CS - 2128

Course Outcomes:

After successful completion of course the students will be able to:

- Describe the meaning of the Object-Oriented Paradigm, and create class hierarchies using theObject-Oriented Design process
- Design and Implement C++ programs for complex problems, making good use of the featuresof the language such as classes, inheritance and templates
- Design and Implement JAVA programs through various Applications / Minor Project
- Learn further through individual investigation of program behavior and debugging efforts andwill develop a professional attitude towards the programming enterprise.

Paper Title: Computer Networks and Security Paper Code : CS – 2116

Course Outcome

After successful completion of course the students will be able to:

- Understand the fundamental concepts of computer networking

- Understand the services of Physical layer, Data Link layer, Network layer, Transport Layer andApplication layer
- Familiarize with significance and techniques of network security, Cryptography, and DataEncryption Standard.
- Understand Web security, Firewalls and Proxy Firewall.

Paper Title: Artificial Intelligence and Soft Computing Techniques Paper Code: CS – 2129

Course Outcome

After successful completion of course the students will be able to:

- Understand various heuristic search techniques and able to apply to various complex problems in Artificial Intelligence.
- Able to analyze and represent knowledge using knowledge representation schemes as Predicate logic and semantic; networks, develop expert systems and natural language processing systems.
- Design and develop solutions to problems using classification and clustering techniques.
- Understand the concept of Fuzzy logic, Genetic algorithms and their applications.

Paper Title : Web Development and Python Programming Paper Code : CS-2130

Course Outcome:

Upon successful completion of this course, students will be able to:

- Design and structure professional web pages using HTML and style them effectively with CSS, including modern layout techniques like Flexbox and Grid.
- Use JavaScript to manipulate the DOM, handle events, and implement basic form validation for interactive web applications.
- Develop Python programs for solving real-world problems by utilizing its data structures, operators, and string manipulation features.
- Apply advanced Python concepts like functions, recursion, file handling, and libraries (e.g., NumPy) to create modular and efficient code.

Paper Title: Analysis and Design of Algorithms Paper Code: CS – 2119

Course Outcomes :

After successful completion of course the students will be able to:

- Analysis of algorithms including recursive algorithms, space and time complexity.
- Understand the Divide and Conquer design strategy to solve different problems like sorting,selection and matrix multiplication.
- The students will also be able to various Dynamic Programming design techniques.
- Backtracking and Branch n Bound design techniques and application of these techniques tosolve problems. Deal with finding lower bounds on complexity for sorting and search problemsusing comparison trees and the lower bounds on parallel computations.NP-Hard and NP-Complete problems and the examples of such problems able be written.

Paper Title: Interactive Computer Graphics Paper Code: CS – 2120

Course Outcomes :

After successful completion of course the students will be able to:

- After completion of course the student will be many basic understanding of the coreconcepts and the structure of an interactive computer graphics and its components andunderstanding of a typical graphics pipeline.
- Perform of geometrical transformations: 2D and 3D using OpenGL to create interactivecomputer graphics.
- A knowledge and understanding of interaction techniques.
- Create interactive graphics applications / minor projects using 'C' programming.

Paper Title: Theory of Computations Paper Code: CS – 2121

Course Outcomes :

After successful completion of course the students will be able to:

- Demonstrate advanced knowledge of formal computation and its relationship to languages
- Distinguish different computing languages and classify their respective types
- Recognize and comprehend formal reasoning about languages.
- Students will demonstrate knowledge of basic mathematical models of computation anddescribe how they relate to formal languages.

Paper Title: Advance JAVA and Network Programming Paper Code: CS – 2122

Course Outcome

After successful completion of course the students will be able to:

- Development of interactive Desktop and Client/Server applications using UI widgets of Swingand RMI
- Access database through Java applications using Java Data Base Connectivity (JDBC)
- Development of dynamic web applications using Servlets and JSP
- Development of Network applications using TCP and UDP sockets

Paper Title: Mobile Communication and Application Development Paper Code: CS – 2123

Course Outcome:

After successful completion of course the students will be able to:

- Understanding of fundamental concepts of GSM, GPRS, CDMA and latest generations ofMobile Telephony
- Familiarization with Mobile database concepts such as hoarding, caching, data delivery,selecting tuning, indexing and various Mobile Operating systems
- Development of interactive Mobile applications using Android platform

- Testing and deployment of Mobile applications on virtual and physical devices

Paper Title: Software Engineering and Project Management Paper Code: CS – 2117

Course Outcome:

After successful completion of course the students will be able to:

- After completion of course the student will have fundamental knowledge of software engineering and will be able to work in diverse application domains.
- Demonstrate an understanding of and apply current theories, models, and techniques that provide a basis for the software lifecycle.
- The student will have ability to use the techniques and tools necessary for engineering practice in software development.
- Apply their foundations in software engineering to adapt to readily changing environments using the appropriate theory, principles and processes.